



GNSS Go User Manual

Version 0.1.2

Global and regional GNSS/CORS station discovery, planning and data access

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This manual describes the current GNSS Go desktop interface and command-line tools. The application loads a bundled station-position snapshot before the first map view and refreshes supported online catalogs silently in the background.

1. Overview

GNSS Go provides one interface for browsing global IGS and regional CORS networks, selecting stations and dates, reviewing a download plan, and retrieving supported GNSS files.

- Global IGS and regional networks across Asia, Europe, the Americas and Oceania.
- OpenStreetMap is preferred when reachable; an offline map is available without network access.
- A bundled station snapshot makes the first map useful immediately; online catalogs update silently afterwards.
- Observation, navigation and precise-product planning/downloading.
- Provider-specific workflows for Japan GEONET and Korea GNSSData.

1.1 Access badges

Badge	Meaning
■	Download can be completed inside GNSS Go, including HTTP/FTP/SFTP or internal browser automation.
■	The station catalog and official source are provided, but the user must complete the provider's own login/download workflow.

2. Windows installation and launch

For end users, install the Windows package from GitHub Releases. For source development/testing, open PowerShell in the repository root:

```
python -m venv .env
.\.env\Scripts\Activate.ps1
python -m pip install --upgrade pip
python -m pip install -e "[build,hatanaka,unix-z]"
gnssgo gui
```

To build standalone GUI and CLI executables locally:

```
python packaging\build.py --clean --gui --cli
```

The output is dist\GNSS-Go.exe for the desktop application and dist\gnssgo.exe for the CLI.

3. Main interface

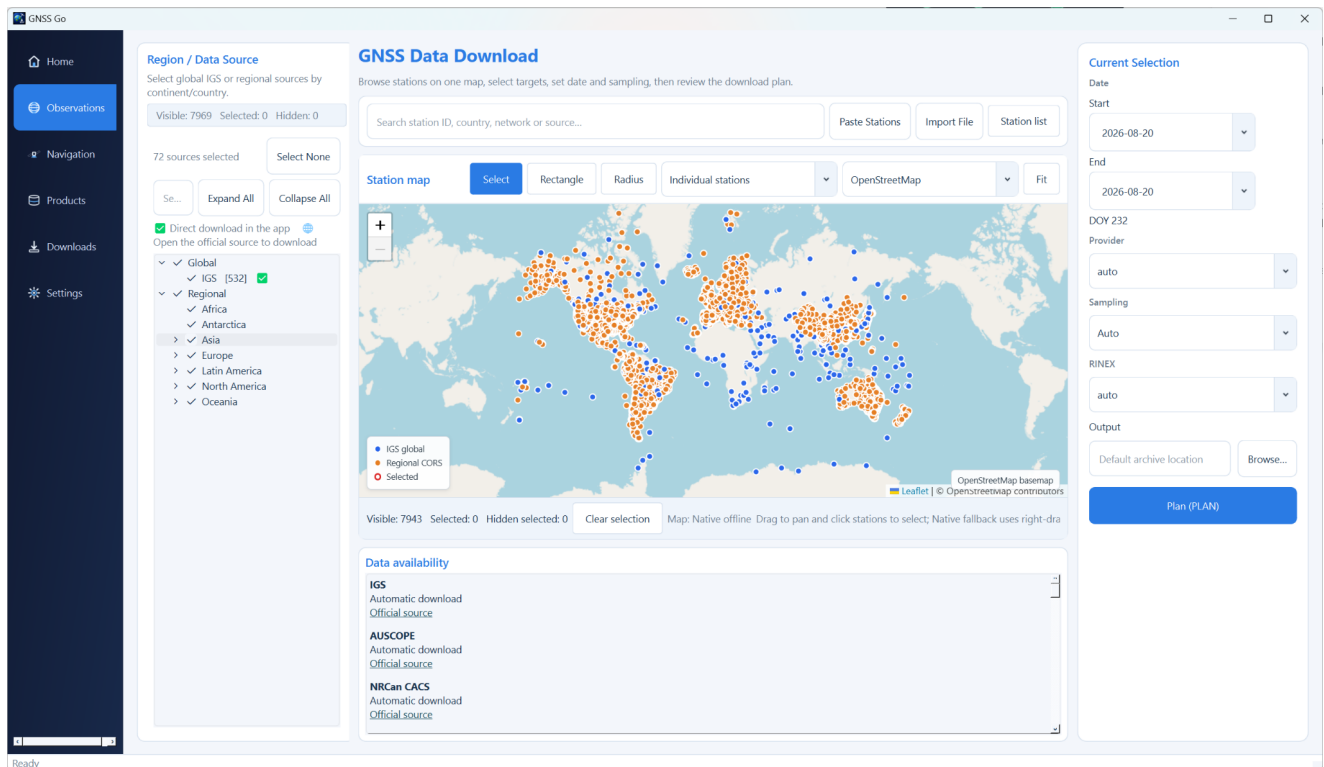


Figure 1. Current GNSS Go English interface (user-supplied screenshot).

The window is organized into the main navigation bar, the region/data-source tree, the station map and availability panel, and the current selection/download configuration panel.

Area	Purpose
Main navigation	Home, Observations, Navigation, Products, Downloads and Settings.
Region / Data Source	Select IGS, continents, countries/regions and integrated networks. Counts reflect the local catalog.
Station map	Click individual stations or use rectangle/radius selection. IGS and regional stations use different marker colors.
Data availability	Shows every selected source; entries are not collapsed into an 'and N more' line.
Current Selection	Configure dates, provider, sampling, RINEX version and output folder.

4. Startup, maps and station catalogs

4.1 Immediate station display

GNSS Go reads the bundled station-position snapshot before constructing the main map. The first view therefore does not depend on live station APIs. When online, supported IGS and regional catalogs refresh silently and merge into the existing catalog without clearing the map first.

4.2 OpenStreetMap

A short connectivity probe runs at startup. If OpenStreetMap is reachable, it is selected as the default basemap; otherwise the application falls back to the bundled offline map.

4.3 Europe

GNSS Go 0.1.2 includes a bundled EPN coordinate fallback so European stations remain visible even when the live EPN catalog is temporarily unavailable. A background EPN refresh can then extend or update the release snapshot.

5. Observation workflow

- Select IGS, a country/region, or a regional network in the left tree.
- Select one or more stations on the map, or use rectangle/radius tools.
- Choose the start and end date in YYYY-MM-DD format.
- Use Provider = auto unless a specific source is required; choose sampling and RINEX as needed.
- Click Plan (PLAN) to review remote files, files already present, files to download and unavailable combinations.
- Start the download and monitor progress in Downloads.

5.1 PLAN versus download

Planning discovers and validates candidate files. One-time server-side download jobs should only be created during the actual download. For Korea GNSSData, GNSS Go creates the temporary ZIP key and consumes it immediately in the same web session.

6. Supported regional CORS data

GNSS Go supports CORS / regional GNSS data for the countries and regions below. This section lists only the supported coverage and whether data can be downloaded directly in GNSS Go or must be obtained from the official provider website.

Region	Direct in GNSS Go	Provider website
Asia	Japan; Korea; Hong Kong	China; Taiwan; Mongolia; Singapore
Europe	France; Germany; Spain; Netherlands; Portugal; Belgium; Greece; Italy; Poland; Romania; United Kingdom; Sweden; Finland; Switzerland; Hungary; Czechia; Slovenia; Ireland; Iceland; Croatia; Norway; Denmark; Estonia; Latvia; Lithuania; Slovakia; Bulgaria; Cyprus; Serbia; Türkiye; Luxembourg; Albania; Bosnia and Herzegovina; North Macedonia; Moldova; Ukraine; Malta; Montenegro	Austria
North America	United States; Canada	-

Region	Direct in GNSS Go	Provider website
Latin America	Brazil; Chile; Mexico; Uruguay	Argentina; Bolivia; Colombia; Ecuador; Peru; Costa Rica; Panama
Oceania	Australia; New Zealand	-
Africa	-	South Africa

7. Navigation, Products and Downloads

7.1 Navigation

Use Navigation for broadcast ephemeris/navigation files. Configure the date and navigation type, review the plan, then download.

7.2 Products

Products covers integrated precise orbit, clock, ERP, IONEX, ANTEX, SINEX and related files. Automatic mode resolves candidates by requested product type, date and configured public providers.

7.3 Downloads

Downloads shows task state, file progress, byte progress, failures and output paths. For partial results, inspect the failed provider/file details before retrying.

8. Settings

- Language: English / Chinese, switchable at runtime.
- Map: OpenStreetMap is preferred when online; the offline map is always available.
- Proxy: Direct, System, HTTP and SOCKS5 modes can be applied to HTTP, FTP and SFTP workflows.
- ChromeDriver: browser-based providers such as Japan GEONET require a compatible Chrome/ChromeDriver environment.
- Download settings: concurrency, retry, resume, automatic extraction and keeping compressed files.

9. Command line

```
gnssgo --help
gnssgo doctor
gnssgo gui
```

The CLI and GUI share the same configuration, providers and download core. Browser-interactive sources are usually easier to operate from the GUI.

10. Troubleshooting

Issue	Action
No online basemap at startup	Check network/proxy settings or switch to the offline map.

Issue	Action
No orange stations in Europe	Version 0.1.2 bundles EPN coordinates. Upgrade and restart if an older cache/release is still in use.
IGS count increases after startup	This is the result of silent background catalog refresh; the bundled snapshot is already usable immediately.
Japan download fails	Verify Chrome/ChromeDriver and use the stage-specific GEONET error message to identify where the workflow stopped.
An old Korea getZip link fails	The temporary key is generated for the live download workflow and should be consumed immediately, not reused later.
PyInstaller is missing	Activate .venv and run <code>python -m pip install -e "[build,hatanaka,unix-z]"</code> .

11. Data and licensing

GNSS Go is an independent data-access client. Copyright, licenses, citation requirements, account requirements and access rules for third-party GNSS/CORS data remain with the original providers. Review the official provider terms before using downloaded data. The GNSS Go repository itself is distributed under the license stated in LICENSE.